

Appendix: Investigating the characteristics of record-breaking (RB) rainfall events in India

A.1 Probability Distributions considered

Here, we provide the details of 8 distributions considered in the study. The suitability of these distributions to the incremental magnitude series (M_j) and the time interval series (L_j) of record events at each j^{th} grid are evaluated using three approaches as mentioned in Section 3.1. The 8 distributions considered are provided in Table T1 below.

Table T1: List of probability distributions considered along with their abbreviations

Sl. No.	Distribution	Abbreviation
1	Generalized Extreme Value Distribution	GEV
2	Generalized Logistic Distribution	GLO
3	Generalized Normal Distribution	GNO
4	Generalized Pareto Distribution	GPA
5	Pearson Type III Distribution	PE3

Also, the quantile $z(F)$ corresponding to each of the distributions for a probability F is provided below. It is to be noted that here the quantile (or variable) is represented as z , however, in this paper, the variables are the incremental magnitudes and the time interval of the record events which are represented as m and t respectively.

Generalized Extreme Value (GEV) Distribution

The GEV distribution finds its application in the analysis of several hydrological data related to extreme events (Martins and Stedinger, 2000). It is a three-parameter distribution with a location (μ), scale (σ), and shape (ξ) parameter. The quantile function of the GEV distribution is given by

$$z(F) = \begin{cases} \mu + \left(\frac{\sigma}{\xi}\right) [1 - \{-\ln(F)\}^\xi] & \text{for } \xi \neq 0 \\ \mu + \sigma \ln\{-\ln(F)\} & \text{for } \xi = 0 \end{cases} \quad (E2)$$

Further details related to GEV distribution and its parameter estimation are provided in Coles (2001), Hosking & Wallis (1997), and Hosking et al. (1985).

Generalized Logistic (GLO) Distribution

The GLO is a three-parameter distribution with a location (μ), scale (σ), and shape (ξ) parameter which has been used widely in several applications. More details related to GLO distribution can be obtained in Kumar & Manju (2022). The quantile function of the GLO distribution is given by

$$z(F) = \mu + \left(\frac{\sigma}{\xi}\right) [1 - \{(1 - F)/F\}^\xi] \quad (E3)$$

Details on estimation of the parameter using the L-moments can be obtained from Hosking (1996) and Hosking & Wallis (1997).

Generalized Normal (GNO) Distribution

The GNO is an extended version of the two-parameter normal distribution that combines (i) lognormal (for positive skewness), (ii) reflected lognormal (for negative skewness), and (iii) Normal distribution (for zero skewness). GNO has three parameters which are location (μ), scale (σ), and shape (ξ) of the distribution. The quantile function of the three-parameter GNO

does not have an explicit form but can be obtained by inverting the probability or the cumulative distribution function (F). The equation provided below demonstrates an indirect way of obtaining the quantile from a GNO distribution.

$$F = \Phi[-\xi^{-1} \ln\{1 - \xi(z - \mu)/\sigma\}]$$

$$z(F) = \Phi^{-1}(F) \quad (E4)$$

where, $\Phi(z) = (2\pi)^{-1/2} \int_{-\infty}^z \exp(-u^2/2) du$ is the standard Normal cumulative distribution function. Details related to GNO and the estimation of its parameters are provided in Hosking (1996) and Hosking & Wallis (1997).

Generalized Pareto (GPA) Distribution

The GPA distribution finds its application to estimate the magnitude of the excess over a threshold corresponding to a probability (F). This distribution is popular to model the peak of the values over a defined threshold which is more popularly known as Peak-Over-Threshold (POT). GPA has three parameters which are location (μ), scale (σ), and shape (ξ) of the distribution. The quantile function of the GPA distribution is given by

$$z(F) = \mu + \sigma\{1 - (1-F)^\xi\}/\xi \quad (E6)$$

The details related to GPA and the estimation of its parameters are provided in Hosking & Wallis (1987), Hosking (1996), and Hosking & Wallis (1997).

Pearson Type III (PE3) Distribution

The PE3 distribution is a combination of Gamma (for positive skewness), reflected Gamma (for negative skewness), and Normal (for zero skewness) distribution. The PE3 is represented by three moments which are: mean (μ), standard deviation (σ), and skewness (ξ). The PE3 has been applied to several hydrological analyses. The quantile function of the PE3 distribution does not have an explicit form, however, it can be obtained indirectly for a corresponding probability (F) as given below

$$F = G\left((z - \mu + 2\sigma/\xi)/\left|\frac{1}{2}\sigma\xi\right|, 4/\xi^2\right); \quad z(F) = G^{-1}(F) \quad \text{for } \xi > 0$$

$$F = 1 - G\left(-(z - \mu + 2\sigma/\xi)/\left|\frac{1}{2}\sigma\xi\right|, 4/\xi^2\right); \quad z(F) = G^{-1}(1 - F) \quad \text{for } \xi < 0 \quad (E7)$$

where, $G(z, \alpha) = \{\Gamma(\alpha)\}^{-1} \int_0^z u^{\alpha-1} e^{-u} du$ is the incomplete gamma integral. More details related to PE3 and the estimation of its parameters are provided in Hosking (1996) and Hosking & Wallis (1997).

A.2 Analysis of the joint distribution of variables using copulas

The joint CDF with S -dimensions of variables can be obtained by utilizing the individual univariate distributions as per the Sklar (1959) theorem (Nelsen, 1998) given as

$$F(z_1, z_2, \dots, z_i, \dots, z_s) = C(F_1(z_1), F_2(z_2), \dots, F_i(z_i), \dots, F_s(z_s)) \quad (E9)$$

where, z_i is the i^{th} variable and $F_i(z_i)$ is the marginal CDF of the variable z_i .

Here, for each j^{th} grid, the incremental magnitude series (M_j) and the time interval series (L_j) are considered simultaneously with $S = 2$.

Therefore, for $S = 2$, Equ. E9 reduces to

$$F(M_j, L_j) = C(F_1(M_j), F_2(L_j)) \quad (E10)$$

$$= C(w, v)$$

where, w is the CDF of the incremental magnitude series and v is the CDF of the time interval series at j^{th} grid.

The copulas considered in this study are provided in Table T2 below

Table T2: List of copulas considered along with their abbreviations

Sl. No.	Copulas	Abbreviation
1	Gaussian Copula	Gaussian
2	Gumble Copula	Gumble
3	Clayton Copula	Clayton
4	Frank Copula	Frank

Gaussian Copula

The Gaussian or Normal Copula is a type of Elliptical Copulas where the dependence between the variables is accounted by using a positive matrix. This copula is found to be suitable when the dimensions are high (Renard and Lang, 2007; Shah and Mishra, 2020).

The Gaussian Copula is expressed as:

$$C_G(w, v) = \Phi_S(\phi^{-1}(w), \phi^{-1}(v)) \quad (E11)$$

where, ϕ is the CDF of standard normal distribution and Φ_S is the CDF of multivariate normal distribution with mean zero and covariance matrix Σ .

Gumbel Copula

The Gumbel copula is first applied to model the joint distribution of the variables where the marginal follow exponential distribution (Gumbel, 1960). It is one kind of the Archimedean which is expressed as

$$C_{GB}(w, v) = \exp\left(-[(-\ln w)^\theta + (-\ln v)^\theta]^{\frac{1}{\theta}}\right) \quad (E12)$$

where, θ is the copula parameter defining the dependence between the variables.

Clayton Copula

Clayton copula is one of the popular kind of copulas from the Archimedean family (Chesneau, 2024). The expression of the Clayton Copula is expressed as

$$C_C(w, v) = [(w)^{-\theta} + (v)^{-\theta} - 1]^{-\frac{1}{\theta}} \quad (E13)$$

Frank Copula

The Frank Copula is the minimal information copula (Sukeda and Sei, 2025) that belong to the Archimedean family (Nelsen, 1998). The joint probability of the two variables with their marginal denoted as w and v is given as

$$C_F(w, v) = -\frac{1}{\theta} \ln\left(1 + \frac{(e^{-\theta w} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1}\right) \quad (E14)$$

A.3 Percentage of grids with no trend

There are arguments regarding the assumption of stationarity of the data for this particular study. Therefore, we have conducted a Mann-Kendall (MK) test (Kendall, 1975) on the data to detect whether the series of the components of the record-breaking (RB) events are stationary or non-stationary. The components of the RB events on which the MK test is applied are the incremental magnitude and the time between RB events. Figure A1 displays that 95% (98%) of the incremental magnitude (time interval) of RB events show no signs of non-stationarity. The outcome of the MK test in the components of the RB events shows that the majority of the series in all grids are stationary, and therefore, the assumption of stationarity is valid in the current work.

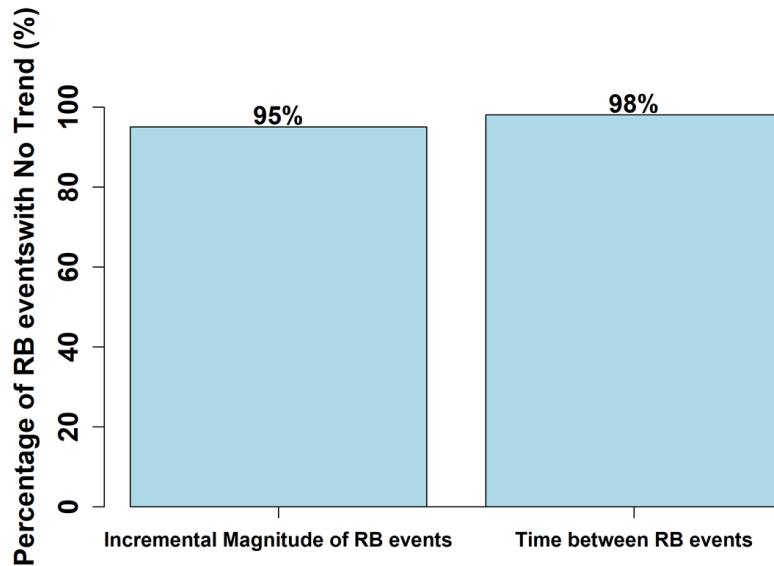


Figure A1 Barplot showing the percentage of grids (all grids with record length > 10) identified with “no trend” using the non-parametric MK test at 0.01 significance level.

However, there could be some confusion that arises regarding the handling of 5% (2%) grids with non-stationarity for incremental magnitude (time interval) of the RB events. We would like to clarify that the percentage of non-stationary grids is very low, and therefore, all the grids can be handled as stationary. Further, since the grids with nonstationarity are very few in number, it can be argued that the non-stationarity observed in those grids may be due to sampling variability in the data and not due to the underlying nonstationarity in the process. For example, the barplot of Figure A2 illustrates the percentage of the 500 simulated series that were falsely identified as having a trend, even though the underlying data were generated without any trend. The bars represent percentages of series showing the false positive rate of trend detection. It means that though the series may be stationary, sampling variability can show a small percentage of data as a non-stationary series. Which means the current study

could also have been a similar issue. Thus, we have considered all the grids to be stationary for the analysis of the components (incremental magnitude and time interval) of RB events.

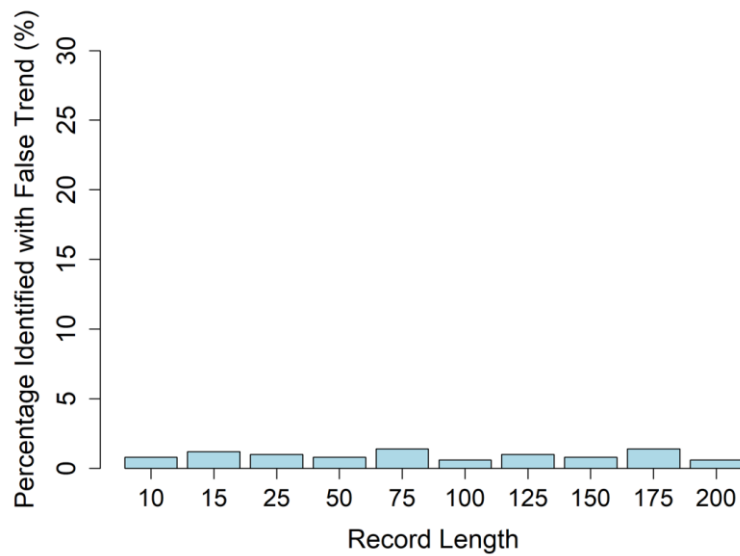


Figure A2 Barplot showing the percentage of grids (from 500 simulations considering no-trend in the series) identified with “trend” using the non-parametric MK test at 0.01 significance level. The time series is simulated with no trend for 500 times, considering different record lengths.

A.4 Boxplot of probability of occurrence of RB events in 10-year time interval

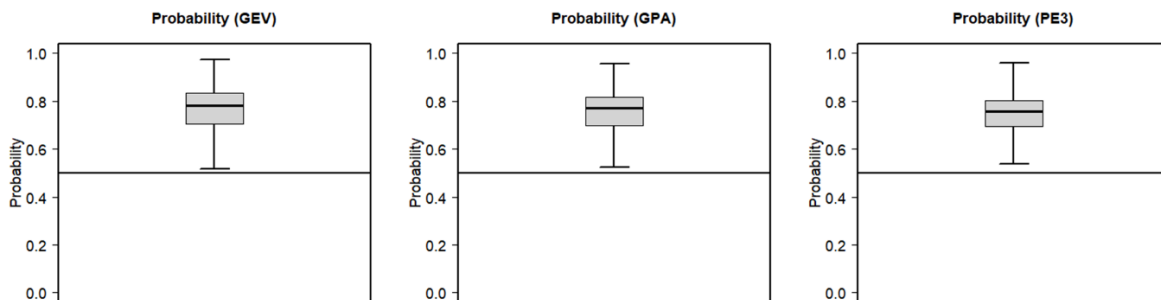


Figure A3 Boxplots of the probability of occurrence of RB events that would occur within ≤ 10 -year time interval across all the grids considered in the study obtained using GEV, GPA, and PE3. For $>95\%$ of the grids, the probability of RB events is > 0.5 .

A.5 Performance of the Gaussian copula in terms of obtaining the joint probabilities of the components of the RB events

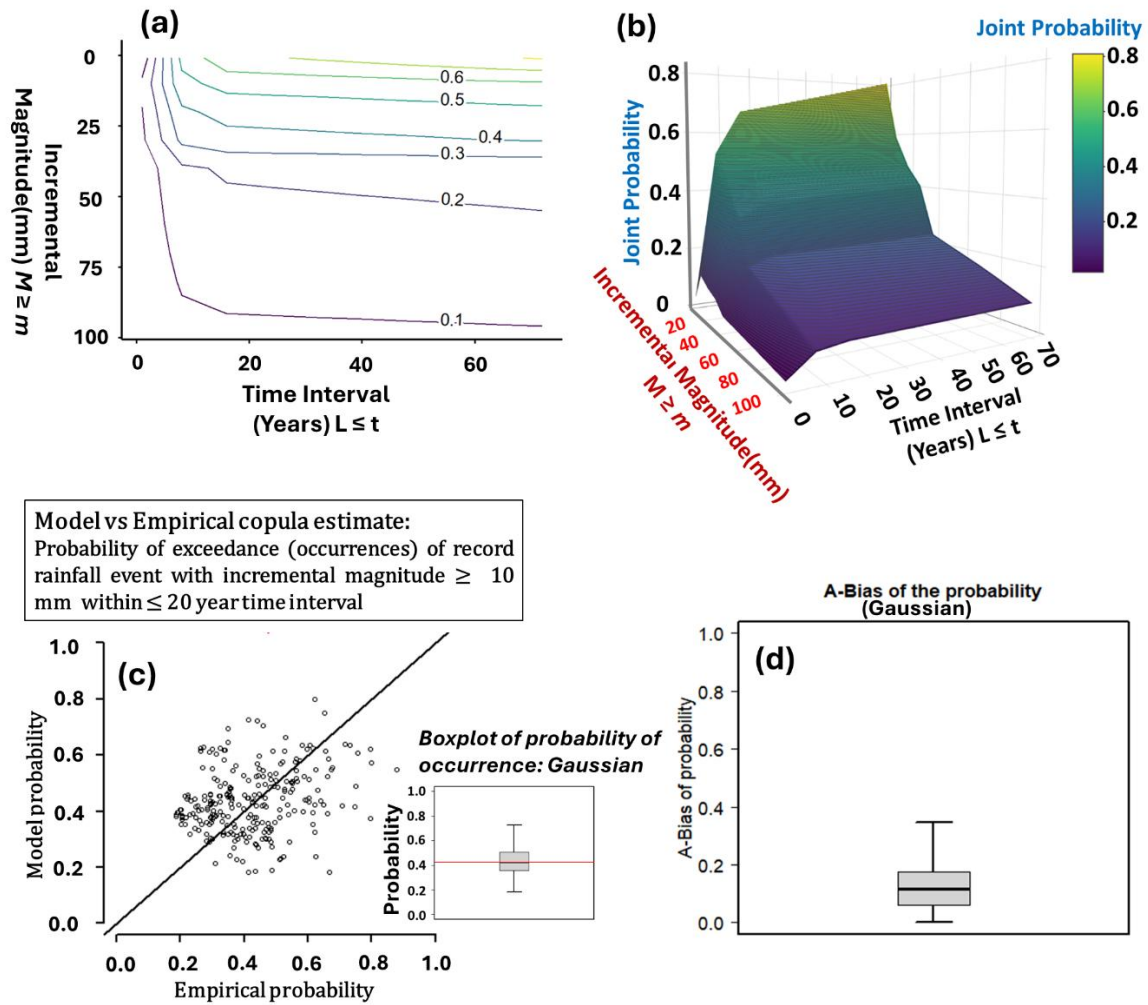


Figure A4 (a) Contours showing the joint probability of RB events with incremental magnitude $\geq$ the given magnitude in a time span of $\leq$ the time interval at a representative grid (79°E and 10.5°N) using the Gaussian copula. (b) 3D representation of the joint distribution of RB events with incremental magnitude $\geq$ the given magnitude in a time span of $\leq$ the time interval at a representative grid (79°E and 10.5°N) using the Gaussian copula. (c) Empirical vs model (Gaussian copula) probability of exceedance for the occurrence of record rainfall events that have an incremental magnitude ≥ 10 mm higher than the previous records and are likely to occur in ≤ 20 years. (d) Absolute bias (A – Bias) in the joint probability of exceedance for incremental magnitude ≥ 10 mm that would occur in ≤ 20 years, obtained from the Gaussian Copula with respect to the actual Empirical Copula.

A.6 Visual comparison between the 2D joint distribution of the components of RB events

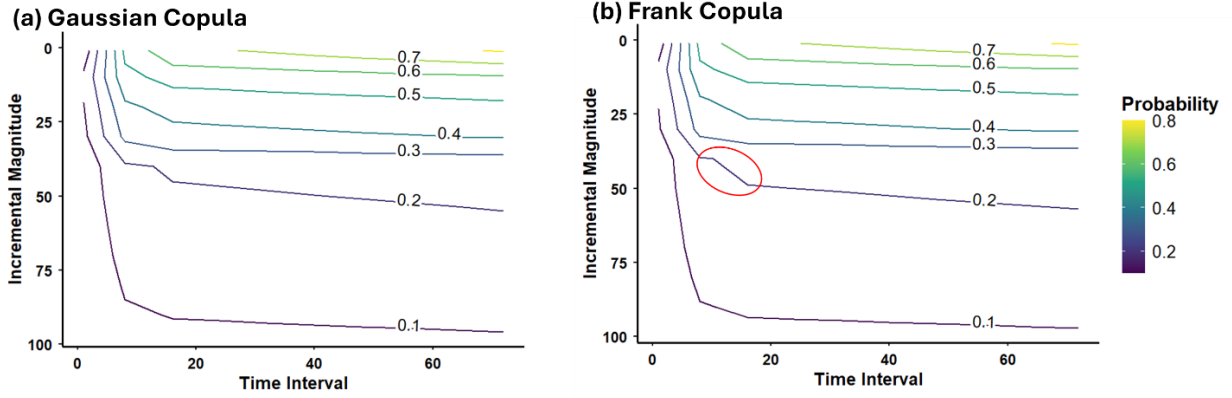


Figure A5 Joint probabilities of the RB events (marked as contours) for incremental magnitude $\geq$ the given magnitude in a time span of $\leq$ the time interval at a representative grid ($79^{\circ}E$ and $10.5^{\circ}N$) using: (a) Gaussian copula and (b) Frank copula. Both the Gaussian and Frank copulas yield similar results except for the location marked by red circle.

A. 7 Justification for using 10 samples for estimation of the parameters

It is to be noted that the problem of adequate sample size is rarely sufficient in environmental applications to enable a reliable estimation of the quantities (Hosking and Wallis 1997). However, the method of L-moment used in the current study for the estimation of the parameters is considered to be robust for small sample (Sankarasubramanian and Srinivasan 1999; Ulrych et al. 2000). Therefore, it is ensured that the L-moment is used for the estimation of the parameters of the distributions. Also, we have therefore attempted to determine the overall distributions that fit the data better across all the grids rather than the individual grids. The idea was to identify an overall suitable distribution for the incremental magnitude and the time interval of the RB events across all the grids, such as the Generalised Extreme Value (GEV) distribution being suitable for the annual maximum series of hydroclimatic variables (AghaKouchak et al. 2013; Coles 2001; Gilleland and Katz 2011).

Therefore, we attempted to investigate how distinct the sensitive at-site L-moments (L-skewness (τ_3^j) & L-kurtosis (τ_4^j)) are from the overall average L-moments i.e., L-skewness (τ_3^{Avg}) & L-kurtosis (τ_4^{Avg}) for the incremental magnitude of the RB events based on the Euclidean distance metric computed for 500 simulation experiments (boxplots in Figure 4 to display the range). Also, we had a closer look at the boxplots of the Euclidean distance of the at-site vs the average L-moments for the 10, 15, and 20 RB events, as shown in Figure A6. It is observed that the median value and the quantile ranges between the 10, 15 and 20 RB events are not very far apart. Table A1 provides the median of the Euclidean distance for GEV, GLO, GNO, GPA, and PE3 distributions across 10, 15, and 20 record lengths, while Table A2 displays the absolute difference in the median Euclidean distance between data of various record lengths for incremental magnitudes of RB events. It is observed from Table A2 that the difference of the median Euclidean distance between the 10 RB events compared to that of 15 (20) RB events varies from 0.04 to 0.06 (0.07 to 0.09) (across different distributions).

Also, a closer investigation of the boxplots of the Euclidean distance between the at-site vs the average L-moments for the 10, 15, and 20 RB events is examined for the time interval of the RB events, as shown in Figure A7. Table A3 provides the median values of the Euclidean distance for GEV, GLO, GNO, GPA, and PE3 distributions for the time interval of RB events

across 10, 15, and 20 record lengths. While, Table A4 shows the absolute difference in the median Euclidean distance between data of various record lengths for the time interval of RB events. It is observed from Table A4 that the difference of the median Euclidean distance between the 10 RB events and that of 15 (20) RB events for the time interval varies from 0.05 to 0.07 (0.07 to 0.1) (across different distributions).

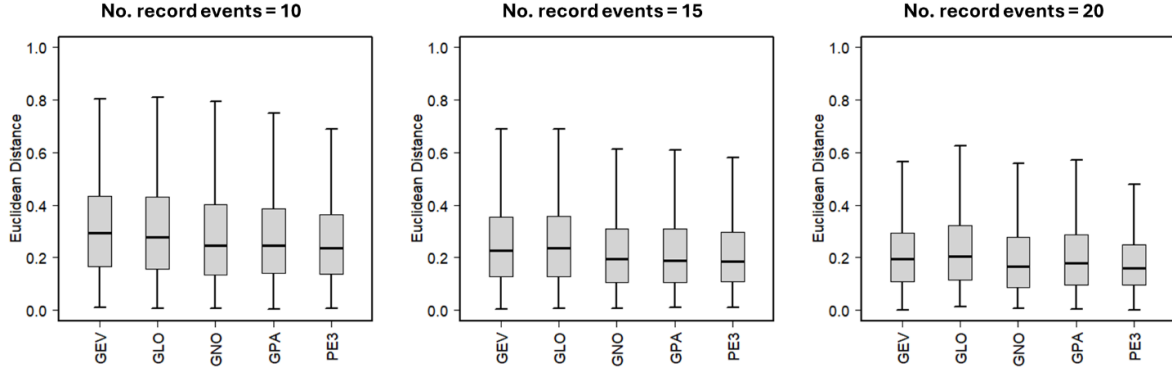


Figure A6 Boxplot of the 500 Euclidean distances (ED_{SA}) between the actual average L -skewness (τ_3^{Avg}) & L -kurtosis (τ_4^{Avg}) (used for generation of data) and the L -skewness (τ_3^{sim}) & L -kurtosis (τ_4^{sim}) obtained from 500 simulated data of **incremental magnitude** considering different distributions (GEV, GLO, GNO, GPA, and PE3) for a number of RB events (10, 15, and 20).

Table A1 Median Euclidean distance between the at-site and the average for the Incremental Magnitude of RB events.

Record length	GEV	GLO	GNO	GPA	PE3
10	0.29	0.28	0.24	0.25	0.24
15	0.23	0.24	0.20	0.19	0.19
20	0.20	0.21	0.17	0.18	0.16

Table A2 Difference in Median Euclidean distance between data of various record lengths for incremental magnitudes of RB events.

Difference in Euclidean distance	GEV	GLO	GNO	GPA	PE3
Between record lengths 10 and 15	0.06	0.04	0.04	0.06	0.05
Between record lengths 10 and 20	0.09	0.07	0.07	0.07	0.08

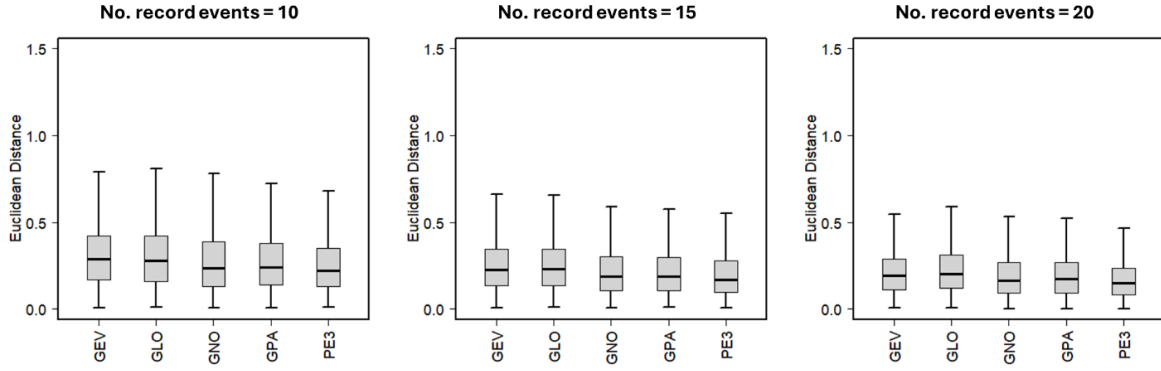


Figure A7 Boxplot of the 500 Euclidean distances (ED_{SA}) between the actual average L-skewness (τ_3^{Avg}) & L-kurtosis (τ_4^{Avg}) (used for generation of data) and the L-skewness (τ_3^{sim}) & L-kurtosis (τ_4^{sim}) obtained from 500 simulated data of **time interval** considering different distributions (GEV, GLO, GNO, GPA, and PE3) for a number of RB events (10, 15, and 20).

Table A3 Median Euclidean distance between the at-site and the average for the time interval of RB events.

Record length	GEV	GLO	GNO	GPA	PE3
10	0.29	0.28	0.24	0.24	0.22
15	0.22	0.23	0.19	0.19	0.17
20	0.19	0.20	0.16	0.17	0.15

Table A4 Difference in Median Euclidean distance between data of various record lengths for the time interval of RB events.

Difference in Euclidean distance	GEV	GLO	GNO	GPA	PE3
Between record lengths 10 and 15	0.07	0.05	0.05	0.05	0.05
Between record lengths 10 and 20	0.10	0.08	0.08	0.07	0.07

Although there is a small deviation in the L-moments, which decides the parameters of the distribution, overall, there is not much difference between the estimates obtained from the 10 RB events and those of the 20 RB events. A record length of 15 to 20 has been used in many studies for estimation of the parameters of the distribution (Hosking and Wallis 1997; O'Brien and Burn 2014; Sankarasubramanian and Srinivasan 1999). Thus, the use of L-moments aids in obtaining the estimates of the parameters with minimal bias for small samples.

A. 8 Justification for using 10 samples for the joint probability of incremental magnitude and time interval of RB events

Concerns may arise regarding the use of limited sample sizes for reliable modelling of the joint probabilities of RB events. However, obtaining sufficiently large sample sizes for robust estimation is a common challenge in environmental and hydroclimatological studies (Hosking and Wallis 1997). To address this issue, additional analyses are performed by fitting the candidate copulas using restricted sample sizes of 10 and 8 RB events across all grids, and the resulting AIC values are compared with those obtained using the actual number of RB events through boxplot analysis, as shown in Figure A8.

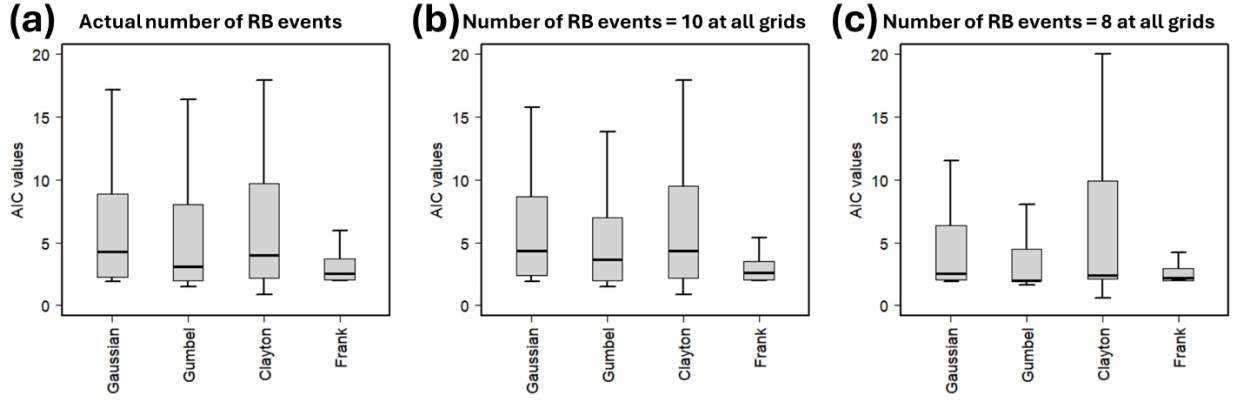


Figure A8 Boxplots of the AIC values obtained for fitting various copulas to capture the joint probability, considering (a) actual RB events, (b) limiting to 10 RB events, and (c) limiting to 8 RB events. The figure displays the Frank Copula to be the most suitable model for capturing the joint probability, as the 5th, 25th, 50th, 75th, and 95th quantiles of the AIC values obtained using the Frank Copula are closer to zero in comparison to the AIC values for other copulas across (a), (b), and (c).

Figure A8 presents three sets of boxplots corresponding to copula fitting by considering: (a) the actual number of RB events, (b) limiting to 10 RB events, and (c) limiting to 10 RB events. The boxplots summarise the AIC distributions through their medians, quantiles, and whiskers. Across all three scenarios, the Frank copula consistently exhibits the lowest 5th, 25th, 50th, 75th, and 95th percentile AIC values, indicating its superior performance irrespective of sample size. Further evidence of the robustness of the Frank copula is reflected in the A-Bias results, where more than 75% of the grids show relatively small errors (≤ 0.154 for RB events ≥ 10 mm occurring within ≤ 20 years). This discussion has now been included in the revised manuscript.